



Course Information Form

Title	Code	Local Credit	ECTS	Lecture (hour/week)	Practical (hour/week)	Laboratory (hour/week)
Mechanics I	MTM3701	4	4	4	0	0

Prerequisite	None
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Semester	Fall
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Course Language	English, Turkish
Level Of Course	First Cycle
Course Type	Required @ Bachelor Programme in Mathematical Engineering
Course Category	Core Courses
Mode Of Delivery	Face-to-Face

Owner Academic Unit	Department of Mathematical Engineering
Course Coordinator	Müslüm Özışık
Instructor(s)	Nazmiye Yahnioğlu, Müslüm Özışık, Ramazan Tekercioğlu
Asistant(s)	

Course Objectives	The objective of the course is teaching to basic knowledge of engineering mechanics. Beginning with basics of vector analysis, students will learn the application of physics and mathematical analysis. At the end of the course, students will learn mechanical systems and then they will make connection between fundamental engineering systems and mathematical means of them.
Course Content	Basic concepts, Scaler, Vector, Unit systems, Newton laws, Force system in the plane, Moment , Particle balance, Plane truss system, Friction, parallel forces and couples, Center of gravity, Frames and machines, flexible cables and chains, force systems in three dimensions, balance of rigid body, Virtual work, Potential energy and stability of equilibrium, Moment of inertia
Recommended Or Required Reading	- 1. R.C. Hibbeler, Fan Sau Cheong, "Statics & Mechanics of Materials SI", Prentice Hall, 2011. - S. Timoshenko ve D.H. Young (çeviren İ. Kayan), Mühendislik Mekaniği, İTÜ, İnşaat Fak. Matbaası, 1990.
Recommended Optional Program Components	None

Course Learning Outcomes

1	Basic engineering knowledge and engineering perspective
2	Engineering problems, problem solving knowledge and skill

Weekly Subjects and Related Preparation Studies

Week	Subjects	Related Preparation
1	Fundamental Concepts, Units of Measurement, Scalars, Vectors, Vector Operations	Related chapter in sources
2	Unit systems, Newton's laws, Force system in the plane,	Related chapter in sources
3	Moment and couple system, equilibrium equations in plane, Equilibrium of particle	Related chapter in sources

4	Simple Trusses, The Method of Joints, The Method of Sections	Related chapter in sources
5	Friction, Friction laws, Wedge friction, belt friction	Related chapter in sources
6	Distributed force in the plane, type of support, Constraints and Degrees of Freedom; The parallel forces system in the plane, The mass and center of gravity, The center of gravity of the compound bodies, Pappus, Guldinus Theorems	Related chapter in sources
7	Frames, Machine	Related chapter in sources
8	Flexible cables, Cables with concentrated Loads. Cables with distributed Loads, Chains	Related chapter in sources
9	Force systems in 3D, equivalent force system	NA
10	Force systems in 3D and its applications	Related chapter in sources
11	Equilibrium of particles in space, Structural analysis	Related chapter in sources
12	Virtual work	Related chapter in sources
13	Potential energy and stable equilibrium	Related chapter in sources
14	Inertia Moment, Moment of inertia of compound bodies, Parallel axis theorem	Related chapter in sources
15	Final exam	Related chapter in sources

Evaluation System		
Activities	Number	Percentage of Grade
Attendance/Participation		
Laboratory		
Application		
Field Work		
Special Course Internship (Work Placement)		
Quizzes/Studio Critics	3	10
Homework Assignments	3	10
Presentations/Jury		
Project		
Seminar/Workshop		
Mid-Terms	1	40
Final	1	40
Percentage of In-Term Studies		60
Percentage of Final Examination		40
TOTAL		100

ECTS Workload Table			
Activities	Number	Duration(Hour)	Total Workload
Course Hours	13	4	52
Laboratory			

Application			
Field Work			
Study Hours Out of Class	13	4	52
Special Course Internship (Work Placement)			
Homework Assignments	3	4	12
Quizzes/Studio Critics	3	2	6
Project			
Presentations / Seminar			
Mid-Terms (Examination Duration + Examination Prep. Duration)	1	2	2
Final (Examination Duration + Examination Prep. Duration)	1	2	2
Total Workload			126
Total Workload / 30(h)			4.20
ECTS Credit			4

Extra Notes	None
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